

Software Implementation and Testing Document

For

Group 1

Version 1.0

Authors:

Matthew C.

Mawi O.

Richard D.

John S.

Hector M.

1. Programming Languages (5 points)

Python

- Reason
 - simple to use, lots of libraries, readable, modern
- Where its used
 - Flask, API routing, database model ORM, user authentication, functionality/logic

HTML/CSS/JS

- Reason
 - simple, fits with Jinja2 (security), and Flask, CSS for customization and JS for functionality
- Where its used
 - Flask templates

SQL

- Reason
 - database operations, schemas
- Where its used
 - Model ORM, user authentication, saving exercises/templates, logging workouts

2. Platforms, APIs, Databases, and other technologies used (5 points)

Platforms

- Flask
 - Used for API routing, model ORM, and functionality

Databases

- SQLite
 - Used for simplified database operations and easy demo/testing

3. Execution-based Functional Testing (10 points)

Functional requirement testing was performed in increment 1 by just running the application locally. In increment 2, pytest will be implemented for a more automated testing workflow.

4. Execution-based Non-Functional Testing (10 points)

Non-functional requirement testing was performed the same way as functional testing; just running the application locally.

5. Non-Execution-based Testing (10 points)

Code reviews were performed before merging each pull request. The reviewer goes through the code and then runs the branch for debugging if necessary, then merges.